SUPERMARKET SALES ANALYSIS

Executive Report | January-March 2024

REVENUE	**Rs 10,035,919**
PROFIT	**Rs 2,820,499**
TRANSACTIONS	**2,500**
AVG. BASKET	**Rs 4,014**

Executive Summary

This report analyses 2,500 cleaned supermarket transactions across three branches. Net sales reached Rs 10,035,919, generating Rs 2,820,499 in profit at an overall margin of 28.1%. Electronics was the leading revenue category, while demand was strongest on Saturday and peaked around 18:00.

The evidence supports focused replenishment, time-based staffing, member bundles, and margin-aware promotions. Because the dataset is simulated and observational, recommendations should be validated with controlled branch experiments.

Prepared as part of the Multi-Domain Data Analysis Portfolio

1. Objectives and Methodology

The analysis addresses four business questions: What sells most? When does demand peak? Which customers and branches contribute most? Which actions can improve revenue and profit?

Data preparation

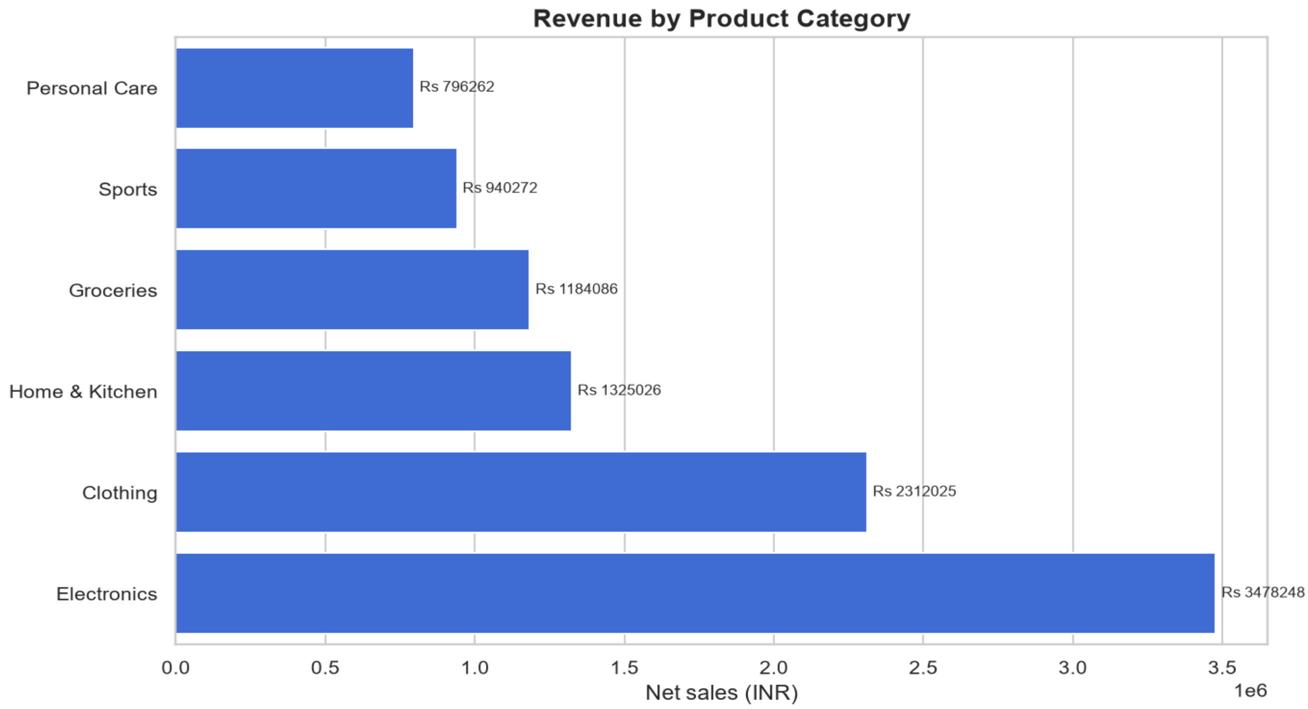
Required columns and numeric ranges were validated. Duplicate invoice IDs were removed, dates and numeric values were standardized, missing ratings were imputed with the median, and unusable values were rejected. Revenue, cost, profit, margin, weekday, month, hour, and time-slot features were then engineered.

Statistical approach

The analysis uses counts, mean, median, quartiles, grouped totals, rolling trends, and Pearson correlation. Mean transaction value captures typical revenue while the median reduces sensitivity to unusually large baskets.

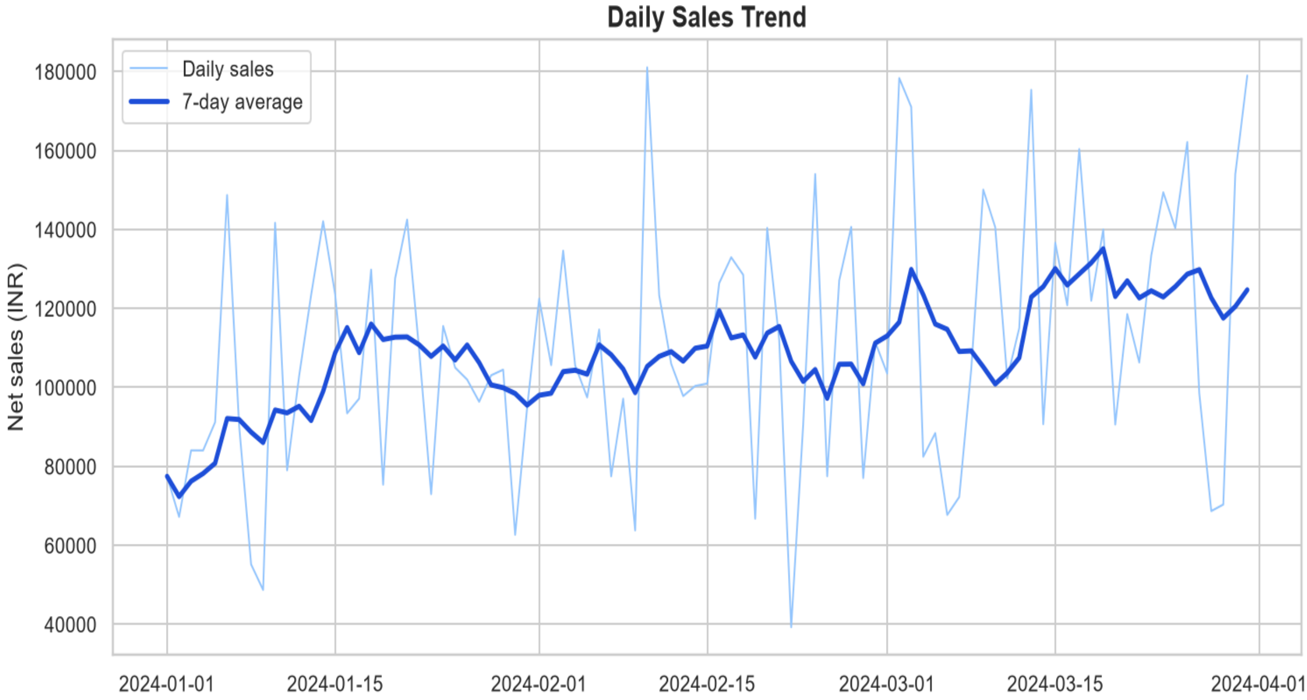
Metric	Value
Average transaction	Rs 4,014
Median transaction	Rs 2,766
Average rating	7.60 / 10
Quantity-sales correlation	0.572
Discount-sales correlation	-0.115

2. Category Performance



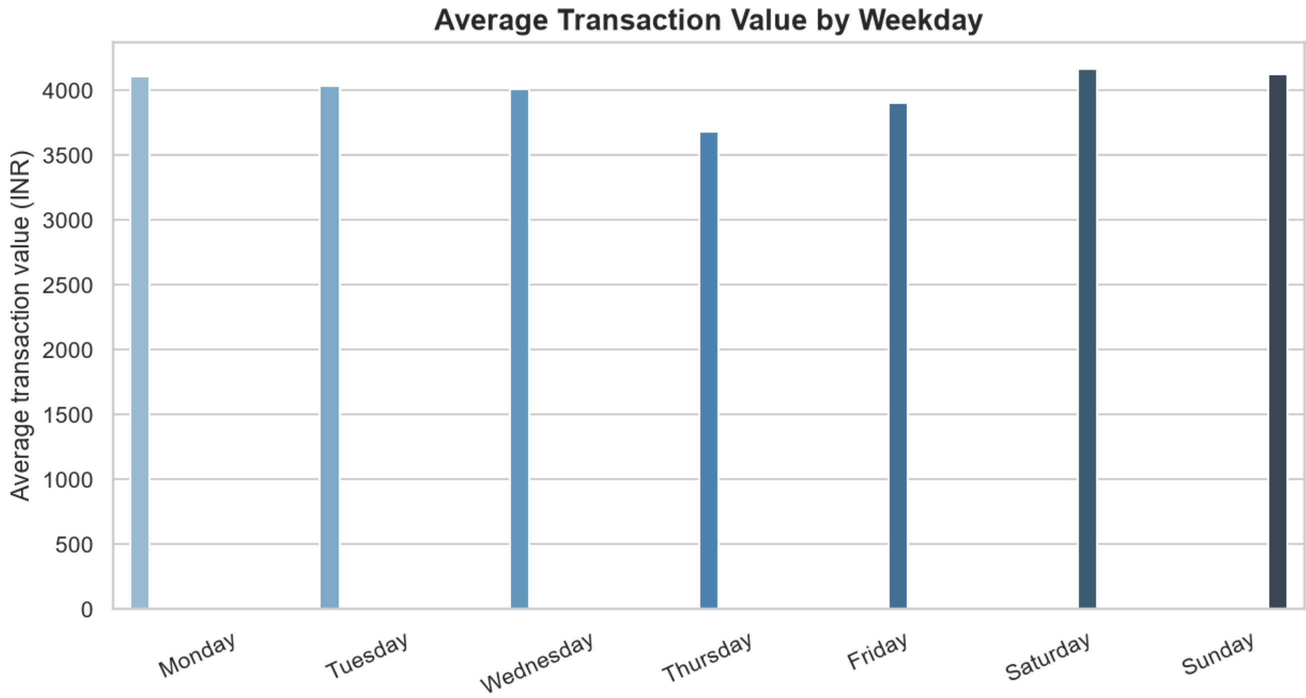
Electronics leads category revenue with Rs 3,478,248. Inventory planning should combine this demand signal with profit margin and stock turnover.

3. Sales Trend



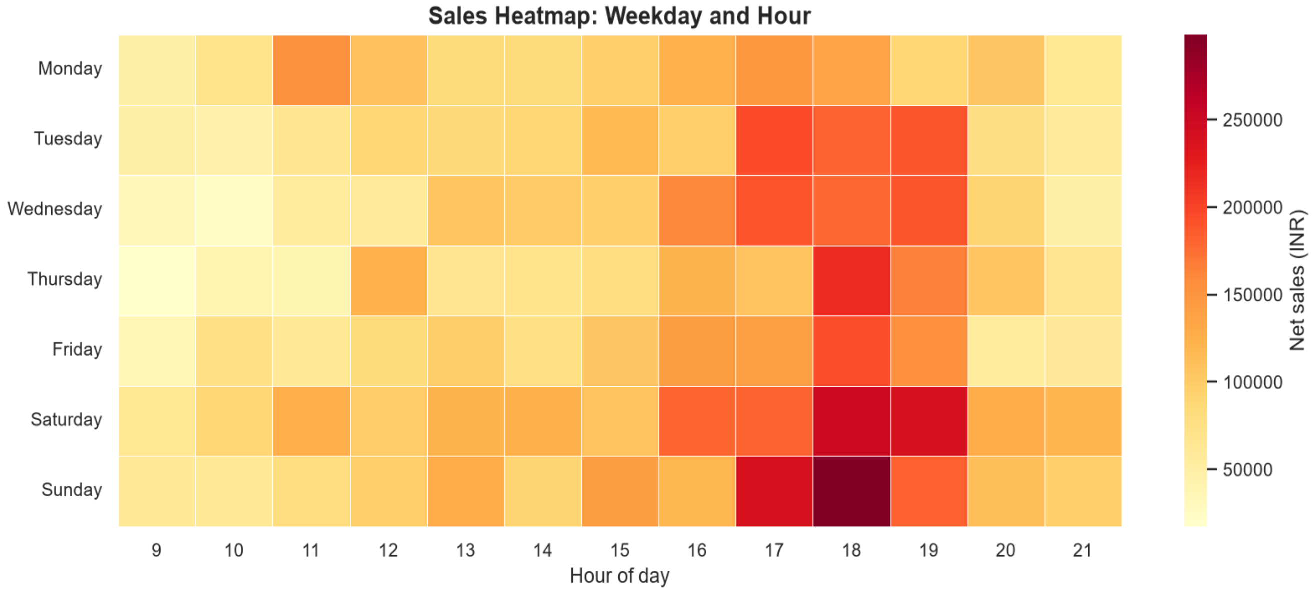
The best individual sales date was 2024-02-10 at Rs 181,030. The seven-day moving average makes the underlying trend easier to distinguish from daily noise.

4. Weekday Performance



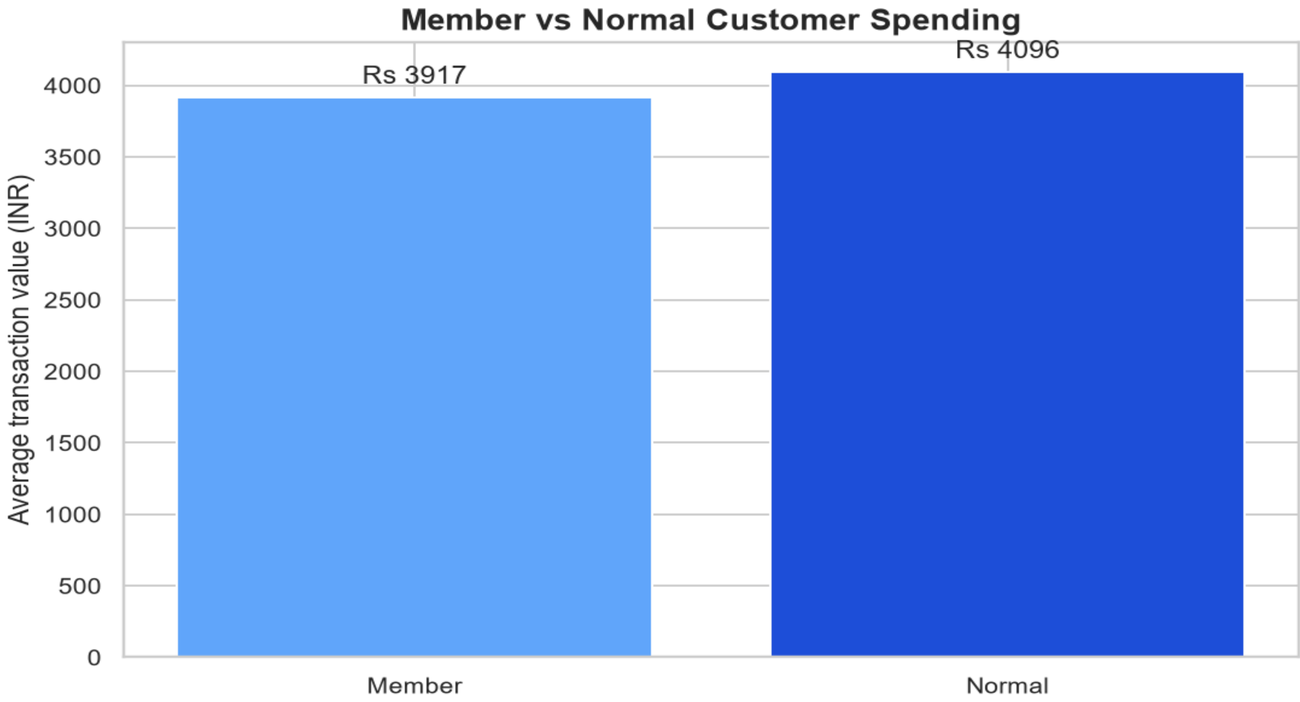
Saturday has the strongest average transaction value. Weekend-oriented bundles and staffing should be tested around this pattern.

5. Peak Shopping Periods



The heatmap identifies the most valuable day-hour combinations. The overall peak is around 18:00, supporting targeted staffing and replenishment before the rush.

6. Customer Behaviour



Member and normal customer results show whether loyalty participation corresponds with higher basket value and ratings. Use member-specific bundles and measure incremental profit, not discount redemption alone.

7. Actionable Recommendations

Inventory: Raise safety stock for leading category-product combinations before peak periods; review weekly sell-through to prevent overstock.

Promotions: Run targeted weekend and evening bundles rather than store-wide discounts; compare incremental profit against a control branch.

Customer retention: Offer member-exclusive cross-category bundles and monitor repeat visits, basket value, and margin.

Operations: Align staff shifts and shelf replenishment with the high-value hour/day cells in the heatmap.

Measurement: Track revenue, gross profit, margin, stock-outs, average basket value, and repeat purchase rate in one weekly scorecard.

Limitations

The dataset is simulated for portfolio practice. Associations may be affected by category mix, branch mix, and seasonality. Correlations are descriptive and should not be interpreted as causal effects.

Conclusion

The project demonstrates an auditable end-to-end retail analytics workflow: validation, cleaning, feature engineering, statistics, visualization, insight generation, testing, and executive reporting.